

Instant Tenant Interconnection: Fabric-in-a-Box for Colocation Facilities

Executive Summary

Tenants expect hyperscaler-level connectivity: instant access to other tenants, sites, and multiple clouds, and features like self-service portals. But building that capability internally requires 12–18 months of development and a dedicated engineering team.

MaiaEdge enables colocation providers to create an automated private connectivity fabric across their facilities in minutes. Providers can deliver instant tenant-to-tenant interconnections, cloud on-ramps, and multi-site connectivity while maintaining end-to-end visibility and ownership of every tenant relationship.

The Challenge

On-Demand Expectations Meet Manual Processes

Tenants expect on-demand connectivity within facilities, across distributed sites, and to cloud on-ramps. Distributed AI is accelerating this demand as tenants require low-latency paths between GPU clusters for training and inference. However, traditional cross-connect processes weren't designed for this speed or flexibility, and building premium connectivity services in-house requires significant operational complexity and infrastructure investment that strains resources and delays time-to-market.

MaiaEdge gives colocation providers a complete fabric-in-a-box solution that transforms data centers into automated interconnection hubs. Providers can instantly connect tenants within the same facility, across campuses, between data centers, to major clouds, and distributed services.

The Solution

Instant Connectivity, Fully Automated

MaiaEdge seamlessly blends Layer 2 Ethernet and Layer 3 DIA connections, enabling data center operators to combine leased lines, dark fiber, and DIA within a unified fabric. API integrations with Equinix and Megaport enable direct cloud access to services like AWS and Azure as a seamless extension of the colocation provider's network, no complex routing protocols required. Providers can federate with partner networks to expand service catalogues while maintaining operational sovereignty and end-to-end visibility across partner circuits. A built-in white-label dashboard lets providers offer tenants a branded self-service portal and end-to-end SLA transparency.

How it Works

Path Border Controller and Path Computation Engine

The MaiaEdge solution combines an edge hardware device, the Path Border Controller (PBC), with an intelligent cloud-native orchestrator, the Path Computation Engine (PCE), to automate path creation across fragmented networks.

At each site, the compact 1RU PBC with dual 100G interfaces pairs with an optional integrated switch that fans out to up to 48 tenant cross-connect ports, delivering a complete turnkey solution. Ports remain inactive until assigned to a tenant in the PCE, ensuring security and control.

The cloud-based PCE continuously computes optimal paths across PBCs. When a tenant requests connectivity, the PCE automatically provisions the logical path with no manual VLAN configuration or routing protocols.

Native integrations with Equinix and Megaport enable automatic connectivity to AWS Direct Connect or Azure ExpressRoute. Providers can deliver cloud access as a seamless extension of their network, retain the invoice and customer relationship, and gain end-to-end visibility and SLA control across partner circuits and cloud on-ramps with hop-by-hop telemetry from customer site to the cloud.

Key Benefits

Automate Manual Cross-Connects

MaiaEdge eliminates ticketing backlogs and manual provisioning. After initial physical cross-connect, services activate instantly through the PCE. What used to require LOAs, meet-me room coordination, and manual configuration taking weeks now happens in minutes. Automation means existing staff can deliver more services faster.

Simplify Cloud Connectivity

Providers can deliver cloud access as a seamless extension of their network. Native API integrations with Equinix and Megaport enable direct connectivity to AWS Direct Connect, Azure ExpressRoute, and other major clouds, without routing complexity.

White-Label Portal

Providers can offer tenants a branded portal for self-service provisioning and hop-by-hop SLA transparency. Tenants can enter cloud profiles, purchase paths to cloud services, and track SLAs from site to cloud. Partner providers remain hidden from the customer view, so tenants only see the colocation provider's brand regardless of path traversal.

DCI Without Routing Complexity

Colocation providers can link their data centers and campuses with simplified data center interconnect (DCI). With MaiaEdge, providers can create a unified fabric across all locations, combining fiber and DIA seamlessly without complex routing infrastructure. Protocol-free forwarding prevents configuration drift and simplifies ongoing operations. Providers can add new sites to the fabric instantly without manual routing configuration or integration projects.

Capabilities and Benefits

Capability	Colocation Benefit
Turnkey fabric hardware	PBC and Integrated Switch provides fabric infrastructure with up to 48 tenant ports per switch
Automated provisioning	Eliminates manual cross-connect configuration after initial physical connection
Fiber & DIA flexibility	Combines both connectivity types in one unified fabric
Protocol-free forwarding	Removes routing complexity, preventing configuration drift
Cloud On-Ramp	Integrations with Equinix and Megaport enable direct access to major cloud providers without routing complexity
Line-rate AES-256-GCM encryption	Secures all paths without performance penalty; meets enterprise security requirements
White-label portal	Provides tenants with a self-service portal and performance insights
Multi-site fabric creation	Connects data centers and campuses without complex routing infrastructure

Business Impact

Unified Network and Automated Operations

Colocation providers can link their data centers and campuses with simplified data center interconnect (DCI). With MaiaEdge, providers can create a unified fabric across all locations, combining fiber and DIA seamlessly without complex routing infrastructure. Protocol-free forwarding prevents configuration drift and simplifies ongoing operations. Providers can add new sites to the fabric instantly without manual routing configuration or integration projects.

Ready to deliver hyperscaler-level connectivity?

Request a demo to see fabric-in-a-box in action.

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